Intermediate Elective

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Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)
library(dplyr)
library(glue)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

```r
extrafont::loadfonts()
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

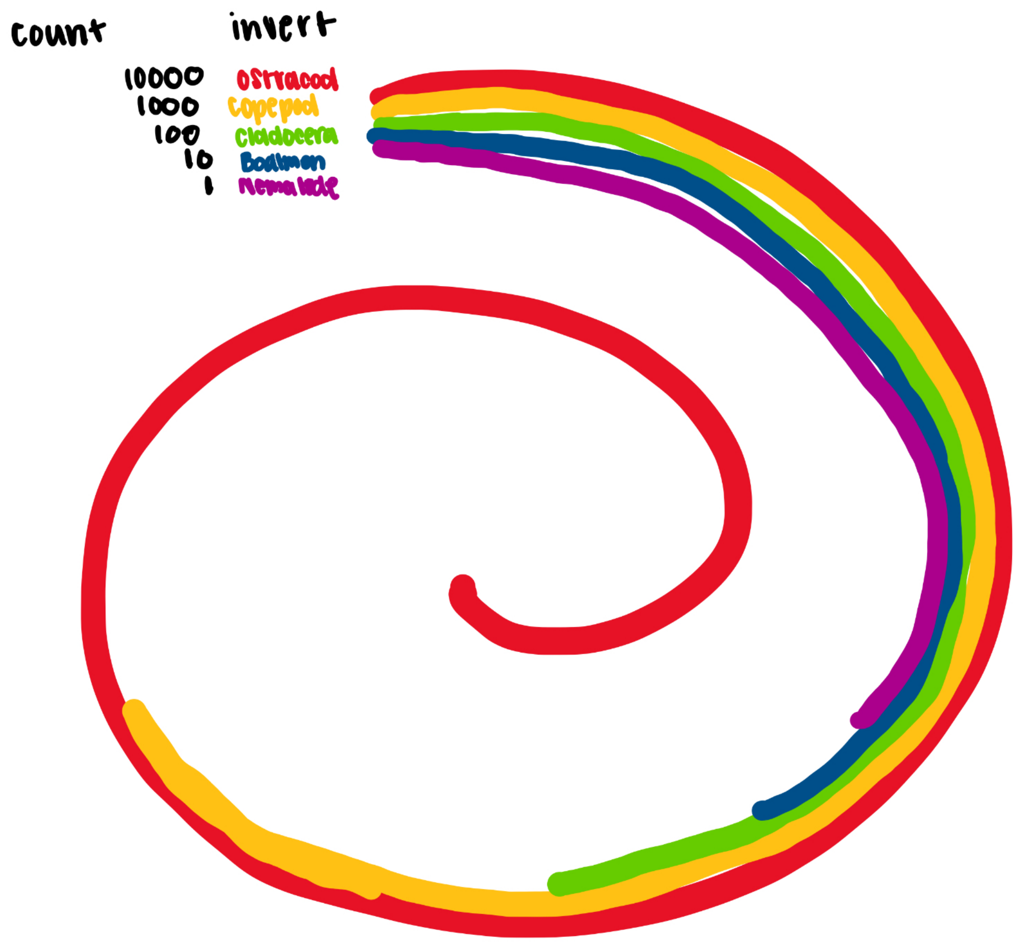
1. Explain your inspiration

- I picked the spiral looking graphic labled 2021_06_dubois_data graphic by Ijeamaka Anyenes.
- This graphic makes sense for my dataset of choice because it is able to display differences in abundance in a fun yet clear and simple way.
- The visualization will show the 5 most abundant inverts during 2024. Each line will represent a different invert and have its own color while the length of the line represents abundance.

2. Plan your figure

- Final visual goal: compare the abundances of the most common inverts found at ncos in 2024

```r
knitr:: include_graphics(here("code","IMG_2970.jpg"))
```



#3. Code your figure

```r
# create new data frame
invert_2024 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  # only include 5 listed invert species sampled in 2024
  filter(year(date) == 2024) |>
  select(ostracod,
         copepod,
         hemiptera_corixidae_boatman,
```

```r
         cladocera,
         nematode) |>
  # change names of columns
  pivot_longer(cols = ostracod:nematode,
               names_to = "taxon",
               values_to = "abundance") |>
  # replace underscores with spaces
  mutate(
    abundance = replace_na(abundance, 0),
    taxon = str_to_title(str_replace_all(taxon, "_", " "))
  ) |>
  # group by taxon
  group_by(taxon) |>
  # add up total abundance for each taxon
  summarise(total_abundance = sum(abundance), .groups = "drop") |>
  # aggrange taxa from highest to lowest abundance
  arrange(total_abundance)
```

```r
# find max x value for the end of the spiral and divid by 1.5 to make the
# spiral wrap tighter
max_x <- max(invert_2024$total_abundance) / 1.5
# calculate the slope to create a spiral shape
slope <- (2-0) / (0 - max_x)
```

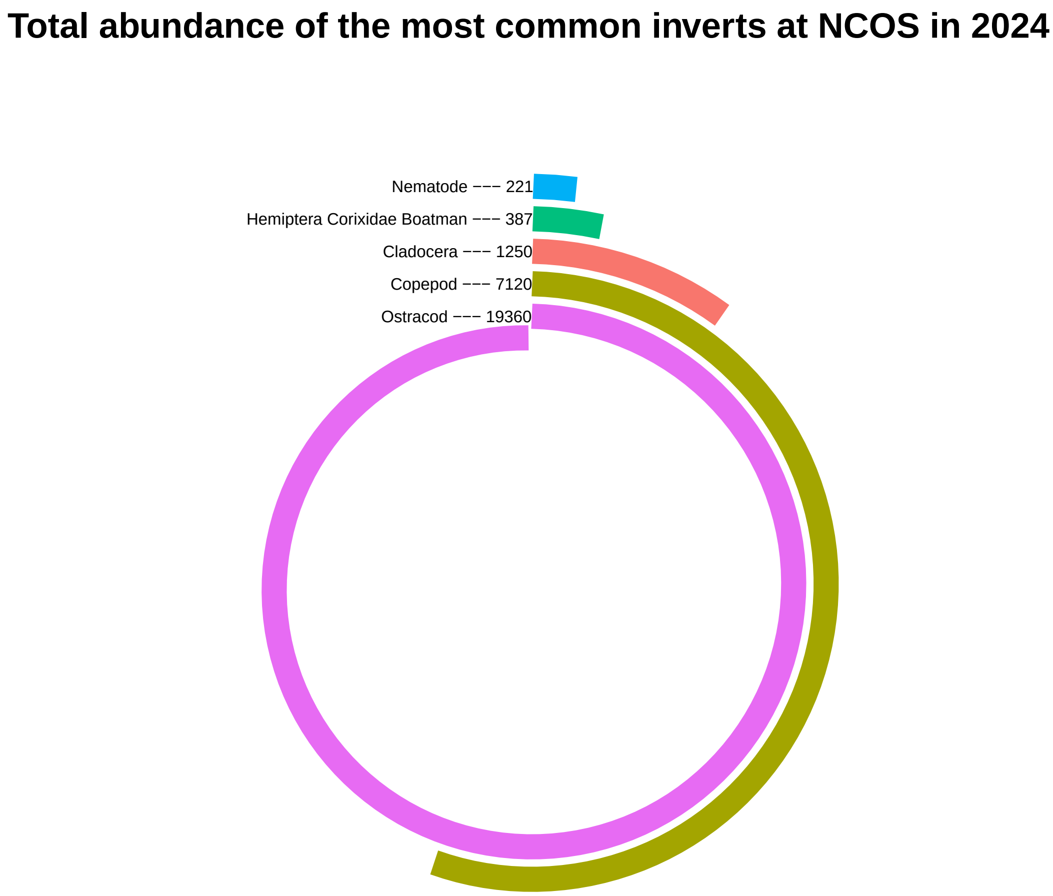
```r
# create new data frame for spiral segments
invert_spiral <- invert_2024 |>
  # give each taxon a different starting y position and separate segments
  mutate(
    taxon = as.factor(taxon),
    y = seq(12, by = -3, length.out = n()),
    # start all segments at x = 0
    x = 0
  ) |>
  # go one taxon at a time
  rowwise() |>
  # the segments either go to taxon abundance or stop at x max
  mutate(
    xend = min(total_abundance, max_x),
    # use slope find the ending y position
    yend = slope * xend + y
  ) |>
  # start second second where the first one ends so it wraps around
```

```r
  mutate(
    y_2 = yend,
    # if abundance is larger than max x create a second segment
    x_2 = 0,
    xend_2 = if_else(total_abundance <= max_x,
                     NA_real_,
                     total_abundance - max_x),
    # calculate ending y position of second segment
    yend_2 = slope * xend_2 + y_2
  )
```

```r
# create new data frame with labels for each taxon
invert_label <- invert_spiral |>
  # keep only columns needed to make labels
  select(taxon, total_abundance, y, x) |>
  # combine taxon name and abundance
  mutate(
    label = glue("{taxon} --- {total_abundance}")
  )
```

```r
# insert spiral style abundance plot
ggplot(data = invert_spiral) +
  # have each segment be a different taxon and the length represent abundance
  geom_segment(aes(x = x, xend = xend,
                   y = y, yend = yend,
                   color = taxon),
               size = 5) +
  # add labels showing taxon name and abundance
  geom_text(data = invert_label,
            aes(x = x, y = y,
                label = label),
            size = 2.5,
            hjust = 1) +
  # add title
  labs(
    title = "Total abundance of the most common inverts at NCOS in 2024"
  ) +
  # convert straight segments to curved ones
  coord_polar(clip = "off") +
  ylim(-25, 15) +
  xlim(-25, max(invert_2024$total_abundance)/1.5) +
  # remove any background markings
```

```r
theme_void() +
# customize plot background and text
theme(
  plot.background = element_rect(fill = "white",
                                 color = NA),
  plot.margin = margin(t = 20, r = 5, b = 5, l = 5),
  panel.background = element_rect(fill = "white",
                                  color = NA),
  plot.title = element_text(hjust = 0.5,
                            face = "bold",
                            size = 15),
  plot.caption = element_text(size = 6),
  # remove legend
  legend.position = "none"
)
```




```r
ggsave(filename = "invert_spiral_plot.png")
```

4. Describe your process

- In order to use someone elses code as a visualization I first ran through their example to try to see what each chunk of code did. I then made a data frame that had the same lay out as the original code. Using this data frame I replaced all of the codes specific to the old data frame with the new code I had made.

- The hardest part of this visualization was it's circularity. I struggled to make the segments bend the way I wanted them to and it was difficult to interpret some of the original code that created the segment slopes.
- Similar to the previous question I think that this plots circularity distinguishes it from anything we have made in class. In class we have mainly focused on the basic plot configurations while this one is less conventional.